INCH-POUND
MIL-DTL-32015/1(AS)
15 October 1999

DETAIL SPECIFICATION SHEET

MICROPHONE, LINEAR BOOM TYPE M-177/AIC

This specification is approved for use by the Naval Air Systems Command, Department of Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-32015 listed in the issue of the Department of Defense Index of Specifications and Standards (DoDDIS) specified in the solicitation..

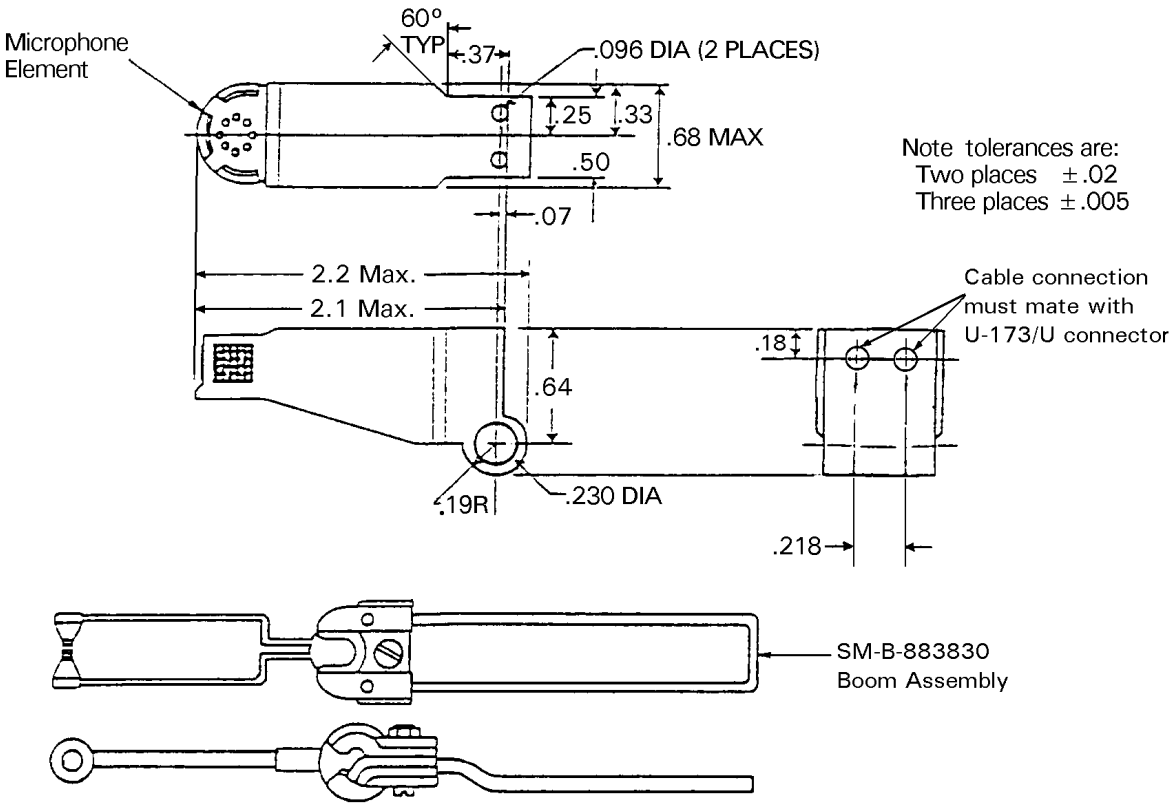

FIGURE 1. Boom microphone assembly

AMSC N/A FSC 5965
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APPLICABLE DOCUMENTS:

MIL-DTL-22442/36	Cable assembly Aircraft Audio

REQUIREMENTS:

DESIGN and CONSTRUCTION: See figure 1.

COMPONENT PARTS: The microphone assembly shall consist of the following:

1 Microphone element
1 SM-B-883830 Boom assembly
1 M22442/36-X Cable assembly

PART or IDENTIFYING NUMBERS (PIN):

M32015/1-1:	With MIL-DTL-22442/36-3 13 inch cable assembly.
M32015/1-2:	With MIL-DTL-22442/36-4 16 inch cable assembly.

WEIGHT: 62g max. for the assembly.

COLOR: Semigloss, lusterless, black.

WIND SCREEN: Each microphone shall be supplied with a wind screen firmly attached.

INPUT CURRENT: 17 mA max. at 27.5 VDC when connected as shown on figure 2 with switch SW1 in the open position

IMPEDANCE: The electrical impedance of the microphone assembly shall be such that the unit shall be capable of meeting the sensitivity and frequency response to a close sound source while connected to a 100 ohm ±1 percent load impedance.

SENSITIVITY: 96.4 mV minimum at 104 dB SPL with 27.5 VDC supply voltage, and shall not vary by more than 1.0 dB when the supply voltage is varied between 22 to 30 volts.

SENSITIVITY AT REDUCED BIAS VOLTAGE: Shall not vary by more than 12 dB at 104 dB SPL from the 27.5 VDC established value when the supply voltage is varied between 12 and 22 volts.

SENSITIVITY AT ALTITUDE: ±3 dB of sensitivity at ground level while at a simulated altitude of 15,000 ft above sea level.

FREQUENCY RESPONSE AT GROUND LEVEL: See figure 3.

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FREQUENCY RESPONSE AT ALTITUDE: Within ±5 dB of the frequency response at sea level, while at a simulated altitude of 15,000 feet above sea level.

NOISE IMMUNITY: See figure 4.

LINEARITY: Resistive load shall be 100 ±1 percent ohms.

GAIN STABILIZATION: Within 100 Ms when supply voltage is between 23 and 30 VDC. Within 150 Ms when supply voltage is between 12 and 22 VDC.

OVERVOLTAGE PROTECTION: 15 minutes operation in the test circuit shown on figure 2 with 27.5 volts applied while the 330 ohm resistor is shorted.

EXPLOSIVE ATMOSPHERE: N/A

ENVIRONMENTAL STRESS SCREENING (ESS): 24 hours of temperature cycling between -20 and +85 °C, with each cycle consisting of 1 hour at each temperature and the change between temperature occurring at a minimum rate of 5 °C per minute. The microphone assemblies shall be functional tested prior to each cold to hot transition and upon completion of the temperature cycling.

MARKING:

M-177/AIC
M32015/1-*
MANUFACTURER'S NAME OR CAGE CODE
CONTRACT NUMBER
DATE CODE
SERIAL NUMBER

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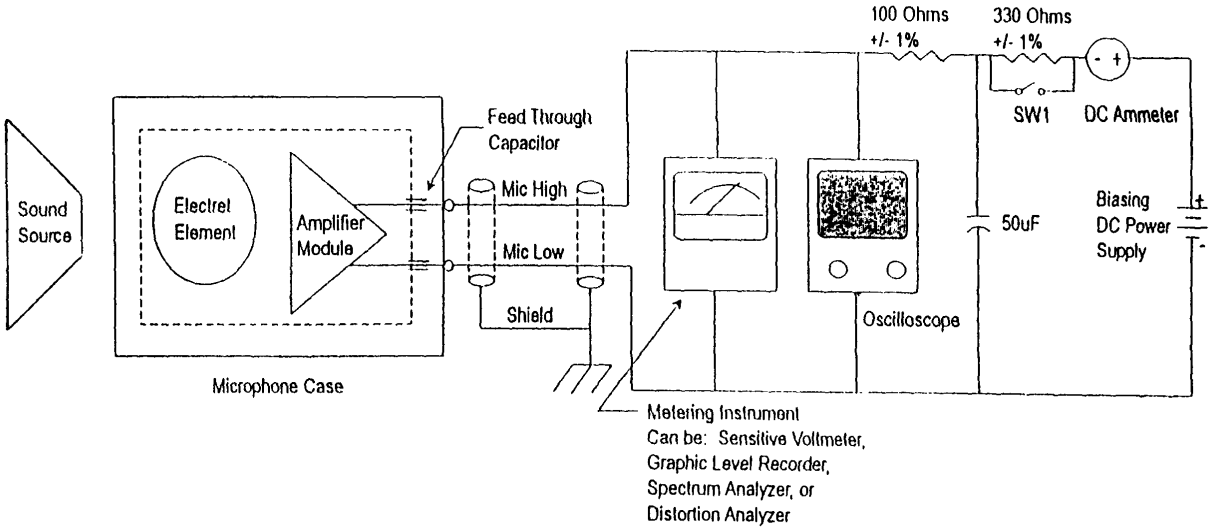

FIGURE 2. Test circuit schematic

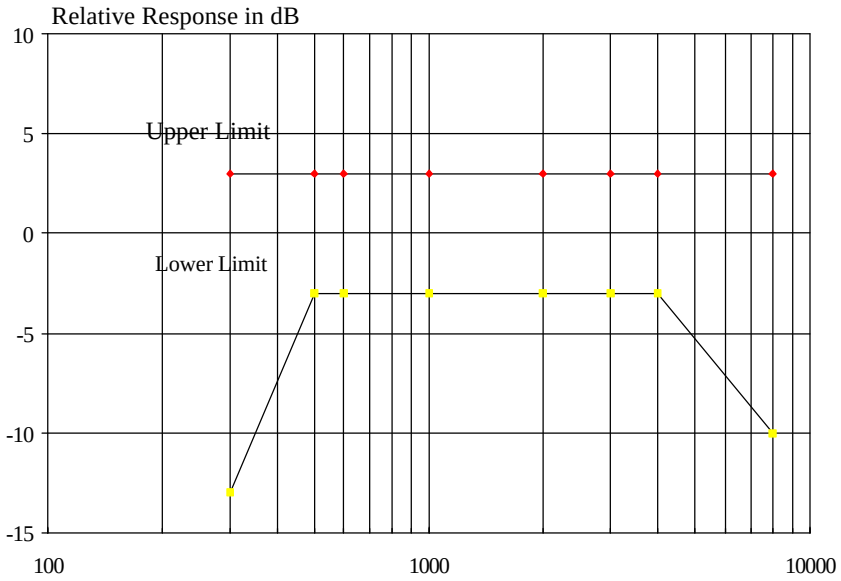

FIGURE 3. Frequency response limit curve.

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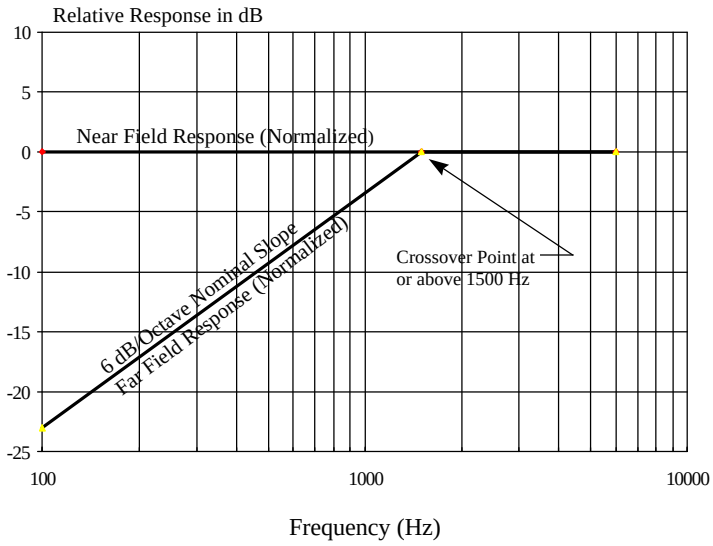

FIGURE 4. Noise immunity characteristics

INTENDED USE: The microphone, M32015/1-* (see PIN), is a noise canceling, dc-powered, high-gain microphone designed for boom mount use on Navy fixed- and rotary- wing helmets. The microphone is intended for use as a part of the airborne communication system which will provide communication of high intelligibility under the extreme noise conditions encountered in military aircraft. The microphone is designed to work with the AIC-14 intercommunication system, specifically control box C-2645D.

Preparing activity:
Navy - AS
(Project 5965-0269-1)

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